

AVNEESH MAHAJAN

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Bhopal, Madhya Pradesh, India

EDUCATION

VIT Bhopal University

Bhopal, India

B.Tech. in Computer Science & Engineering (Specialization in AI & ML) | CGPA: 7.43/10.0 Expected 2027

Coursework: Deep Learning, NLP, CV, RL, Linear Algebra, Statistics, OS, DSA

TECHNICAL SKILLS

- **Machine Learning & AI:** PyTorch, TensorFlow, Keras, OpenCV, YOLOv5, ONNX, RL, Pandas, Polars, NumPy, Scikit-learn
- **DevOps & Cloud:** Docker, Docker Compose, AWS (EC2, S3, RDS, Lambda, VPC, IAM, API Gateway)
- **Languages:** Python, C#, SQL, JavaScript, TypeScript
- **Backend & Data:** FastAPI, AsyncIO, OAuth2, PostgreSQL, MongoDB, Redis, RabbitMQ, Celery, pgvector, SQLAlchemy

EXPERIENCE

Cognifyz Technologies

Remote

Machine Learning Intern

Jan 2026 – Feb 2026

- Developed a predictive regression model using **Scikit-learn** to forecast restaurant ratings, optimizing feature selection algorithms to maximize model accuracy.
- Engineered a content-based recommendation system, personalizing restaurant suggestions by analyzing structured user preferences and cuisine categorical data.
- Leveraged **Polars** for high-performance geospatial data processing, significantly reducing data cleaning and pipeline execution latency.

PROJECTS

Eventsnap — Async Facial Recognition API [\[backend\]](#) [\[frontend\]](#)

Python, FastAPI, ONNX, pgvector

- Architected a distributed, horizontally scalable microservice for processing high volumes of event photos.
- Engineered a stateless GPU inference container utilizing **ONNX Runtime** and InsightFace to efficiently compute 512-dimension vector embeddings.
- Implemented sub-millisecond **Cosine Similarity** facial matching across large datasets by leveraging **PostgreSQL pgvector**.
- Orchestrated asynchronous background encoding pipelines using Celery and RabbitMQ to decouple image processing from the FastAPI backend.

CAR_TRACK — RL Environment Engineering

Python, Physics Engine, WebSockets

- Engineered a custom 2D physics-based simulation environment from scratch, specifically designed as a training ground for **Reinforcement Learning** (RL) agents.
- Designed complex state-space generators, implementing mathematical polygon intersection and ray-casting logic to simulate **LiDAR sensors** and calculate agent rewards.
- Architected a real-time FastAPI backend utilizing WebSockets for high-frequency, bi-directional telemetry streaming between the simulation client and external RL algorithms.

Catalyst — Data Preprocessing & Analytics Platform [\[github\]](#)

Python, FastAPI, Pandas, SQLAlchemy

- Developed an automated data preprocessing service designed to ingest, clean, and analyze messy datasets for downstream Machine Learning pipelines.
- Built scalable data aggregation endpoints using **FastAPI** and asynchronous SQL queries (SQLAlchemy) to serve analytics to the frontend.
- Implemented data normalization and missing-value imputation workflows to accelerate feature engineering and exploratory data analysis.

CERTIFICATIONS

Applied Machine Learning in Python [\[certificate\]](#)

University of Michigan (Coursera)

AWS Academy Graduate — Machine Learning Foundations [\[badge\]](#)

Amazon Web Services

AWS Academy Graduate — NLP [\[badge\]](#)

Amazon Web Services

AWS Academy Graduate — Cloud Foundations [\[badge\]](#)

Amazon Web Services